

● HARD

● ALGEBRA

Algebra

10 questions · ~15 min

03

Q1

GRID-IN ALGEBRA

The line with the equation $\frac{4}{5}x + \frac{1}{3}y = 1$ is graphed in the xy-plane. What is the x-coordinate of the x-intercept of the line?

STUDENT-PRODUCED RESPONSE

--	--	--	--

.	/	.	.
---	---	---	---

0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q2

ALGEBRA

The linear function g is defined by $gx = b - 15x$, where b is a constant. If $gc + 7 = \frac{c}{4}$, where c is a constant, which of the following expressions represents the value of b ?

- A. $\frac{15c}{4}$
- B. $\frac{19c}{4} + 7$
- C. $\frac{61c}{4} + 105$
- D. $15c + 105$

Q3

ALGEBRA

Ken is working this summer as part of a crew on a farm. He earned \$8 per hour for the first 10 hours he worked this week. Because of his performance, his crew leader raised his salary to \$10 per hour for the rest of the week. Ken saves 90% of his earnings from each week. What is the least number of hours he must work the rest of the week to save at least \$270 for the week?

- A. 38
- B. 33
- C. 22
- D. 16

Q4

GRID-IN ALGEBRA

$$3(kx + 13) = \frac{48}{17}x + 36$$

In the given equation, k is a constant. The equation has no solution. What is the value of k ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q5

GRID-IN ALGEBRA

$$\frac{2}{5}x + \frac{7}{5}y = \frac{2}{7}$$

$$gx + ky = \frac{5}{2}$$

In the given system of equations, g and k are constants. The system has infinitely many solutions. What is the value of $\frac{g}{k}$?

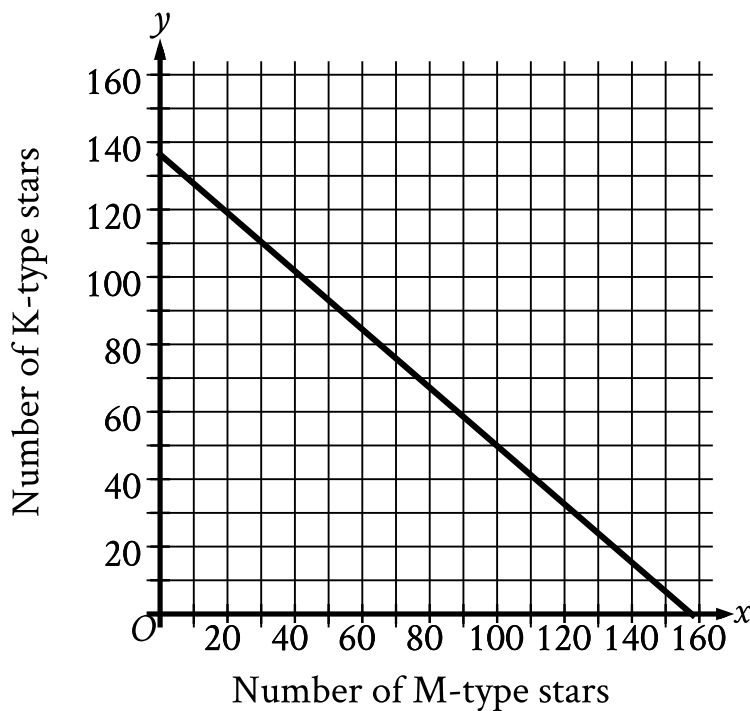
STUDENT-PRODUCED RESPONSE

--	--	--	--

.	/	.	/	.
---	---	---	---	---

0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



- The line slants gradually down from left to right.
- The line passes through the following approximate points:
 - (0 comma 136)
 - (158 comma 0)

A certain open star cluster contains M-type stars and K-type stars. The estimated total mass of M-type and K-type stars in this open star cluster is 127,882 quettagrams. The graph shown models the possible combinations of the number of M-type stars, x , and K-type stars, y , that could be in this open star cluster if all the M-type stars have the same estimated mass and all the K-type stars have the same estimated mass. Based on the graph, which of the following is closest to the estimated mass, in quettagrams, of each M-type star in this cluster?

- A. 811
- B. 938

C. 51,904

D. 75,978

Q7

ALGEBRA

The function $f(x)$ is defined as 19 more than 4 times a number x . If $y = f(x)$ is graphed in the xy -plane, what is the best interpretation of the x -intercept?

- A. When $f(x) = 0$, the number is $-\frac{19}{4}$.
- B. When the number is 0, $f(x) = 19$.
- C. The value of $f(x)$ increases by 1 for each increase of 4 in the value of the number.
- D. For each increase of 1 in the value of the number, $f(x)$ increases by 4.

$$\begin{aligned}y &> 2x - 1 \\ 2x &> 5\end{aligned}$$

Which of the following consists of the y-coordinates of all the points that satisfy the system of inequalities above?

A. $y > 6$

B. $y > 4$

C. $y > \frac{5}{2}$

D. $y > \frac{3}{2}$

Q9

ALGEBRA

A factory makes 9-inch, 7-inch, and 4-inch concrete screws. During a certain day, the number of 9-inch concrete screws that the factory makes is 5 times the number n of 7-inch concrete screws, and the number of 4-inch concrete screws is 22. During this day, the factory makes 100 concrete screws total. Which equation represents this situation?

- A. $95n + 7n + 422 = 100$
- B. $9n + 7n + 4n = 100$
- C. $5n + 22 = 100$
- D. $6n + 22 = 100$

Q10

ALGEBRA

$$2x + 3y = 7$$

$$10x + 15y = 35$$

For each real number r , which of the following points lies on the graph of each equation in the xy -plane for the given system?

- A. $\frac{r}{5} + 7, -\frac{r}{5} + 35$
- B. $-\frac{3r}{2} + \frac{7}{2}, r$
- C. $r, \frac{2r}{3} + \frac{7}{3}$
- D. $r, -\frac{3r}{2} + \frac{7}{2}$